FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: transformer-z-tolerance-envelope-v29
Model status: CALCULATED
Captured UTC: 2026-11-07T18:20:41.746Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: projectName | Entered value: 12 kV ? 208 V ? Worked example
Input path: tolerancePercent | Entered value: 7.5
Input path: buses | Entered value: 3
Input path: transformers | Entered value: 1
Input path: faultType | Entered value: 3phase
Input path: scenario | Entered value: base

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: CALCULATED
Calculated output path: method | Value: transformer-z-tolerance-envelope-v29
Calculated output path: version | Value: 29.0.0
Calculated output path: tolerancePercent | Value: 7.5
Calculated output path: transformerCount | Value: 1
Calculated output path: searchedCases | Value: all 2 +/- tolerance combinations
Calculated output path: scope | Value: Selected-fault results from the existing Studio network engine, repeated with each enabled transformer nameplate %Z scaled by (1 - tolerance) or (1 + tolerance); X/R is kept. Per bus, the largest and smallest results over the nameplate case and the searched combinations are reported with the governing combination. Lower %Z does not always raise every bus current: with generator or motor contributions, a stiffer transformer can divert their current, so the extremes are searched rather than assumed.
Calculated output path: limitations[0] | Value: Applies one common tolerance magnitude; tested impedance from a certified test report supersedes this envelope.
Calculated output path: limitations[1] | Value: Extremes are searched at tolerance corners only; with mixed sources a true extreme can occasionally lie between corners (difference is typically well below the tolerance effect).
Calculated output path: limitations[2] | Value: Does not change utility maximum/minimum strength settings; use Study setup scenarios for source variation.
Calculated output path: limitations[3] | Value: Not an ANSI C37 or IEC 60909 equipment-duty certification and not a PPE or arc-flash determination.
Calculated output path: reference[0] | Value: IEEE C57.12.00: two-winding transformer impedance tolerance of +/-7.5 % of nameplate (user-selectable here).
Calculated output path: reference[1] | Value: Eaton Bussmann point-to-point method: multiply transformer %Z by 0.9 for a worst-case available fault current.
Calculated output path: reference[2] | Value: NFPA 70 (NEC) 110.24(A): field-marked maximum available fault current with the date the calculation was performed.
Calculated output path: issues | Value: []
Calculated output path: buses[0].busId | Value: bus-bac65be4-ad4e-41e4-bded-057e2a88552a
Calculated output path: buses[0].busName | Value: Utility / POI
Calculated output path: buses[0].voltageV | Value: 12000
Calculated output path: buses[0].nominal.status | Value: CALCULATED
Calculated output path: buses[0].nominal.totalA | Value: 3400.0000000000005
Calculated output path: buses[0].nominal.xr | Value: 8.000000000000002
Calculated output path: buses[0].nominal.peakA | Value: 8143.109400350162
Calculated output path: buses[0].maximumWithTolerance.status | Value: CALCULATED

Calculated output path: buses[0].maximumWithTolerance.totalA | Value: 3400.0000000000014
Calculated output path: buses[0].maximumWithTolerance.xr | Value: 8
Calculated output path: buses[0].maximumWithTolerance.peakA | Value: 8143.109400350164
Calculated output path: buses[0].maximumWithTolerance.governingCase | Value: 12 kV / 208 V transformer: -7.5 %
Calculated output path: buses[0].minimumWithTolerance.status | Value: CALCULATED
Calculated output path: buses[0].minimumWithTolerance.totalA | Value: 3400.0000000000005
Calculated output path: buses[0].minimumWithTolerance.xr | Value: 8.000000000000002
Calculated output path: buses[0].minimumWithTolerance.peakA | Value: 8143.109400350162
Calculated output path: buses[0].minimumWithTolerance.governingCase | Value: nameplate
Calculated output path: buses[0].maximumIncreasePercent | Value: 2.220446049250313e-14
Calculated output path: buses[0].minimumDecreasePercent | Value: 0
Calculated output path: buses[1].busId | Value: bus-17b44807-0915-47e6-8bc7-ae3131a3a269
Calculated output path: buses[1].busName | Value: 208 V transformer secondary
Calculated output path: buses[1].voltageV | Value: 208
Calculated output path: buses[1].nominal.status | Value: CALCULATED
Calculated output path: buses[1].nominal.totalA | Value: 32725.85775961685
Calculated output path: buses[1].nominal.xr | Value: 8.000000000000014
Calculated output path: buses[1].nominal.peakA | Value: 78379.48234025245
Calculated output path: buses[1].maximumWithTolerance.status | Value: CALCULATED
Calculated output path: buses[1].maximumWithTolerance.totalA | Value: 34907.10397060491
Calculated output path: buses[1].maximumWithTolerance.xr | Value: 7.999999999999999
Calculated output path: buses[1].maximumWithTolerance.peakA | Value: 83603.63720059802
Calculated output path: buses[1].maximumWithTolerance.governingCase | Value: 12 kV / 208 V transformer: -7.5 %
Calculated output path: buses[1].minimumWithTolerance.status | Value: CALCULATED
Calculated output path: buses[1].minimumWithTolerance.totalA | Value: 30801.17917752866
Calculated output path: buses[1].minimumWithTolerance.xr | Value: 8.000000000000018
Calculated output path: buses[1].minimumWithTolerance.peakA | Value: 73769.81520658922
Calculated output path: buses[1].minimumWithTolerance.governingCase | Value: 12 kV / 208 V transformer: +7.5 %
Calculated output path: buses[1].maximumIncreasePercent | Value: 6.665207149068775
Calculated output path: buses[1].minimumDecreasePercent | Value: 5.881216609280793
Calculated output path: buses[2].busId | Value: bus-3f52b4cc-4f8d-4227-a6fd-d22193cd0075
Calculated output path: buses[2].busName | Value: Main panel
Calculated output path: buses[2].voltageV | Value: 208
Calculated output path: buses[2].nominal.status | Value: CALCULATED
Calculated output path: buses[2].nominal.totalA | Value: 16706.17335282207
Calculated output path: buses[2].nominal.xr | Value: 1.266220695569668
Calculated output path: buses[2].nominal.peakA | Value: 26264.644475141395
Calculated output path: buses[2].maximumWithTolerance.status | Value: CALCULATED
Calculated output path: buses[2].maximumWithTolerance.totalA | Value: 17172.445067428267
Calculated output path: buses[2].maximumWithTolerance.xr | Value: 1.2229569887423921
Calculated output path: buses[2].maximumWithTolerance.peakA | Value: 26818.687532828295
Calculated output path: buses[2].maximumWithTolerance.governingCase | Value: 12 kV / 208 V transformer: -7.5 %
Calculated output path: buses[2].minimumWithTolerance.status | Value: CALCULATED
Calculated output path: buses[2].minimumWithTolerance.totalA | Value: 16260.23367548939
Calculated output path: buses[2].minimumWithTolerance.xr | Value: 1.3089355272786227
Calculated output path: buses[2].minimumWithTolerance.peakA | Value: 25733.025633135745
Calculated output path: buses[2].minimumWithTolerance.governingCase | Value: 12 kV / 208 V transformer: +7.5 %
Calculated output path: buses[2].maximumIncreasePercent | Value: 2.7910144636888568
Calculated output path: buses[2].minimumDecreasePercent | Value: 2.669310726728147
Calculated output path: nec11024Label.title | Value: MAXIMUM AVAILABLE FAULT CURRENT
Calculated output path: nec11024Label.equipment | Value: MDP-1
Calculated output path: nec11024Label.voltage | Value: 208
Calculated output path: nec11024Label.availableFaultA | Value: 17172.445067428267
Calculated output path: nec11024Label.availableFaultText | Value: 17.17 kA

Calculated output path: nec11024Label.xr | Value: 1.2229569887423921
Calculated output path: nec11024Label.basis | Value: Transformer impedance reduced by 7.5 % (worst case)
Calculated output path: nec11024Label.dateOfCalculation | Value: 2026-11-07
Calculated output path: nec11024Label.calculatedBy | Value:
Calculated output path: nec11024Label.reference | Value: NEC 110.24(A) field marking. Re-calculate and re-mark when modifications change the available fault current (NEC 110.24(B)).

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: transformer-z-tolerance-envelope-v29
Evidence / limitation: Timestamp | Value: 2026-11-07T18:20:41.746Z
Evidence / limitation: outputs.scope | Value: Selected-fault results from the existing Studio network engine, repeated with each enabled transformer nameplate %Z scaled by (1 - tolerance) or (1 + tolerance); X/R is kept. Per bus, the largest and smallest results over the nameplate case and the searched combinations are reported with the governing combination. Lower %Z does not always raise every bus current: with generator or motor contributions, a stiffer transformer can divert their current, so the extremes are searched rather than assumed.
Evidence / limitation: outputs.limitations[0] | Value: Applies one common tolerance magnitude; tested impedance from a certified test report supersedes this envelope.
Evidence / limitation: outputs.limitations[1] | Value: Extremes are searched at tolerance corners only; with mixed sources a true extreme can occasionally lie between corners (difference is typically well below the tolerance effect).
Evidence / limitation: outputs.limitations[2] | Value: Does not change utility maximum/minimum strength settings; use Study setup scenarios for source variation.
Evidence / limitation: outputs.limitations[3] | Value: Not an ANSI C37 or IEC 60909 equipment-duty certification and not a PPE or arc-flash determination.
Evidence / limitation: outputs.reference[0] | Value: IEEE C57.12.00: two-winding transformer impedance tolerance of +/-7.5 % of nameplate (user-selectable here).
Evidence / limitation: outputs.reference[1] | Value: Eaton Bussmann point-to-point method: multiply transformer %Z by 0.9 for a worst-case available fault current.
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